

# **23CSE301**

## **Machine Learning Capstone**

### **Phase - I Review**

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**Paper ID:** 045

**Paper Title:** Improving Phishing Email Detection Performance Through Deep Learning With Adaptive Optimization

# Understanding the Problem:

## Problem Statement:

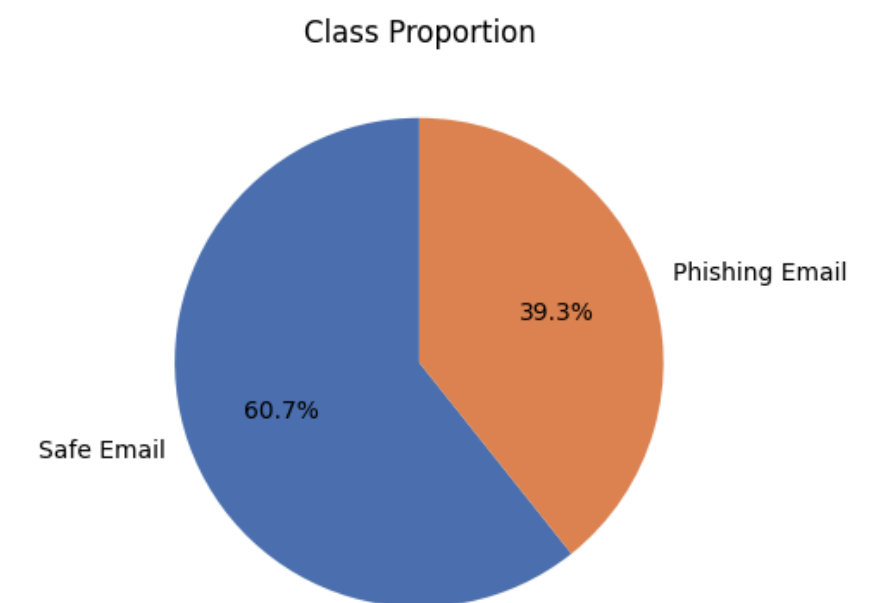
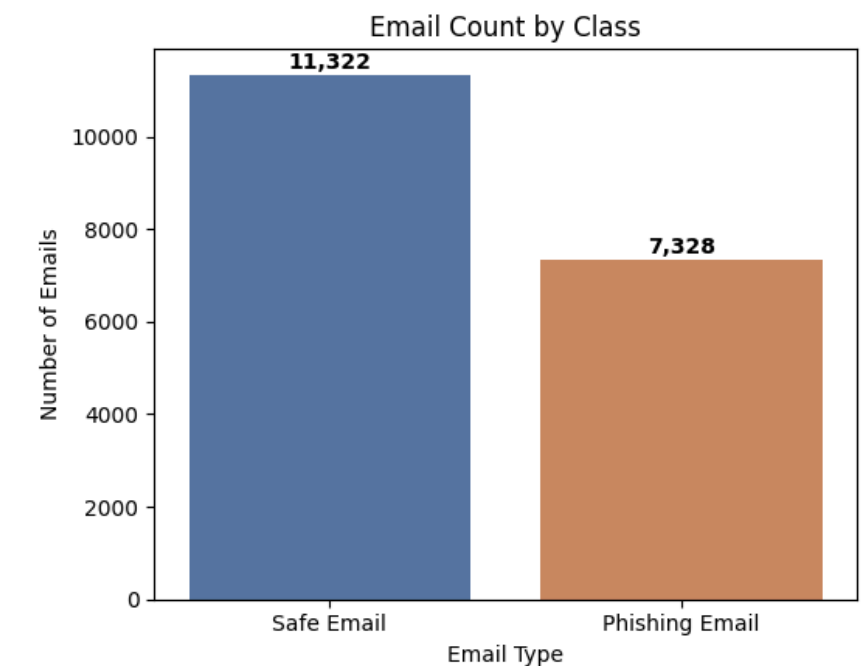
- Phishing attacks are growing more sophisticated, closely imitating trusted sources and placing a heavy burden on cybersecurity.
- This makes it difficult for both users and traditional filters to tell legitimate messages from malicious ones.
- Traditional filters rely on known keywords and patterns, so they often fail against new or cleverly worded phishing attempts.
- This gap creates the need for a smarter, context aware detection system.

## Interpretation:

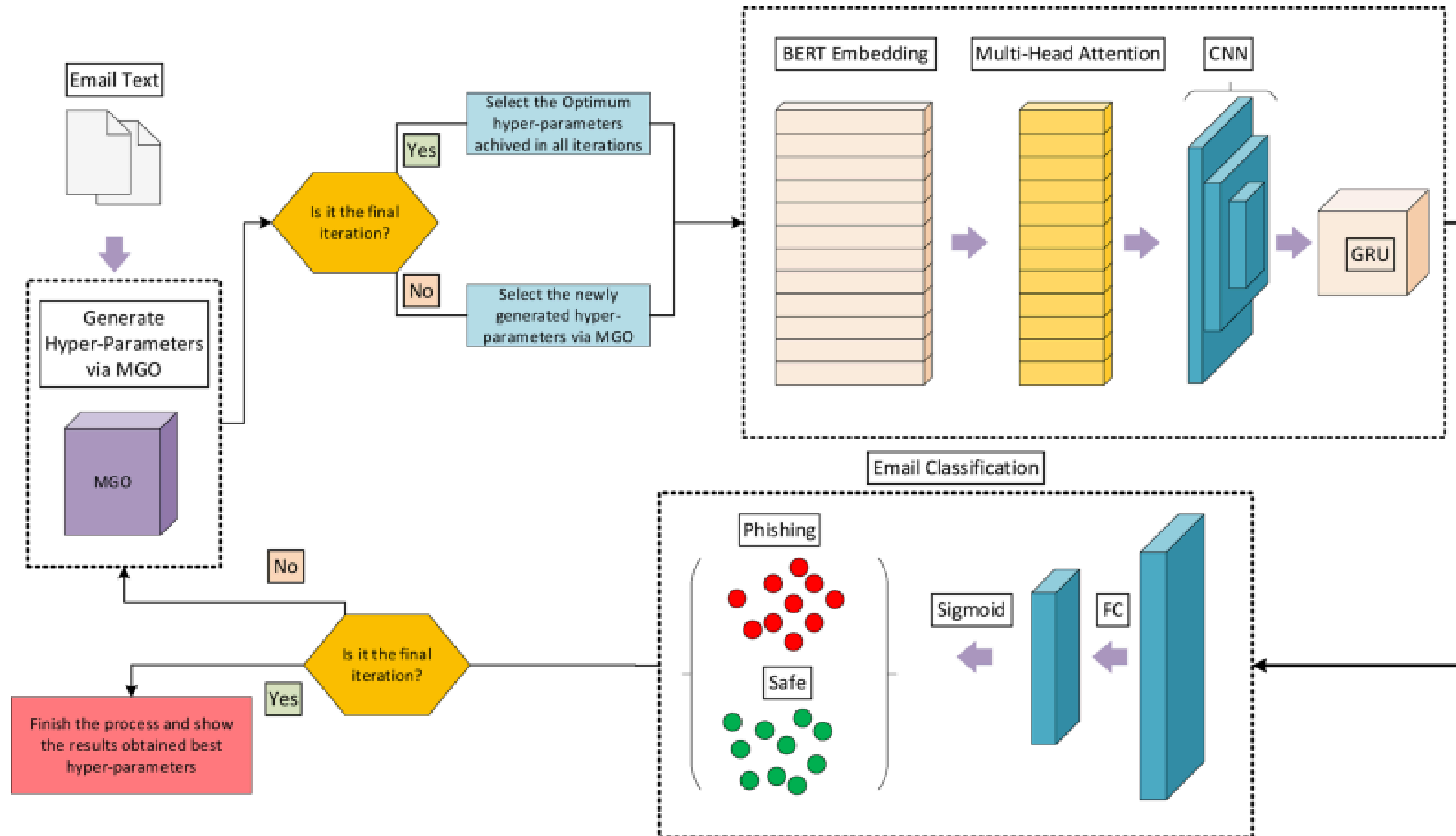
- The problem can be modeled as a **Supervised Learning** problem, since the dataset provides labeled examples (Phishing / Safe).
- It is a **Classification** problem, and falls under **Binary Classification**.
- It is a **Deep Learning problem**, since detecting phishing requires understanding context, not just matching keywords.
- To catch phishing, we need to **understand the context behind an email** and what it is really trying to say, not just the words it uses.

# Dataset Description & Interpretation:

- URL: <https://www.kaggle.com/datasets/subhajournal/phishingemails/data>
- Shape: 18,650 rows x 3 columns
- Total No. of Samples: **18,650**
- Total No. of Columns: **3**
- Column Headers: **['Unnamed: 0', 'Email Text', 'Email Type']**
- Distribution of Safe vs Phishing Emails in the Dataset used:
  - Safe Emails - **11,322 (60.71 %)**
  - Phishing Emails - **7,328 (39.29 %)**
- It is a **mildly imbalanced** classification problem. As **the dataset is a bit skewed towards Safe Emails (60.71%)** Compared to Phishing Emails (39.29%).



# Interpretation of Approach Used:



## Key Terms:

- Tokenization
- MGO (Mountain Gazelle Optimizer)
- BERT (Bidirectional Encoder Representations from Transformers)
- Multi-Head Attention
- CNN (Convolutional Neural Network)
- GRU (Gated Recurrent Unit)

Source: The above image has been extracted from the base paper.